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DATE	30 th November 2009
VERSION	v0.7

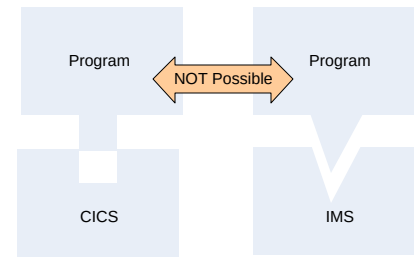
J2EE Application Server Cost Reduction

1. Background

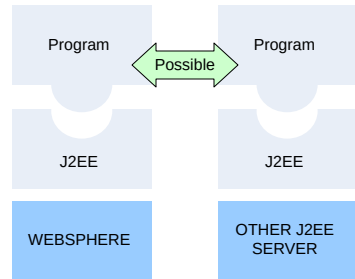
Websphere 5.1 Now out of support from IBM. This means we will have to upgrade to newer Websphere version to receive support from vendor. This upgrade will require a significant migration effort. We can use this opportunity to evaluation costs and quality of product and compare against other options available.

Program Portability:

CICS/IMS: Migration of program code is not possible between platforms

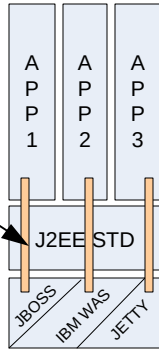


J2EE: Java language standards allow migration of program code to competing vendor platforms



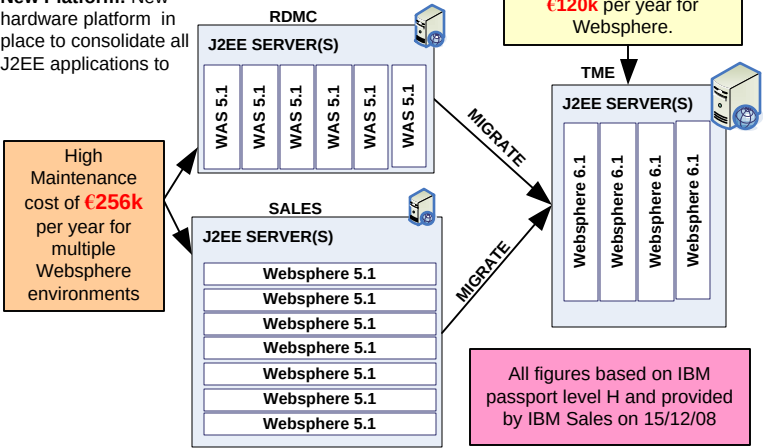
VENDOR SPECIFIC FEATURES: All J2EE vendors have extra functionality specific to their product which can be used by developers.

PROBLEM: If vendor features used then migration difficult

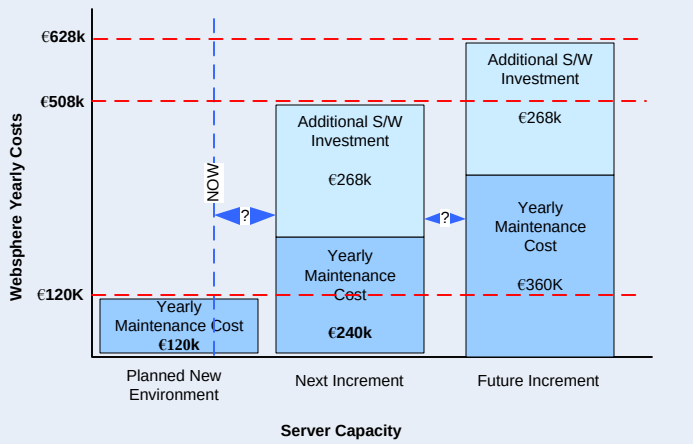


2. Current Condition

New Platform: New hardware platform in place to consolidate all J2EE applications to

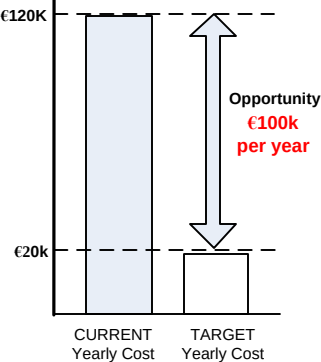


Cost Scaling: Websphere maintenance costs scale on every server installed DEV, UAT & PROD. There is also a cost to acquire increased licence.



3. Evaluation

Target Setting: We have set a yearly cost target of 100k per year

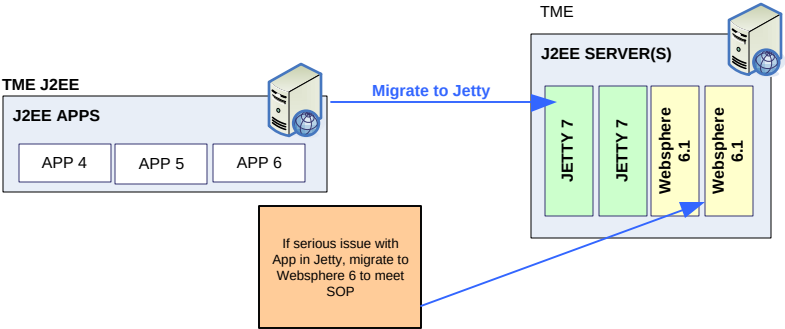


Evaluation Summary Details of options evaluated.

Option	VENDOR SUPPORT QUALITY	JAVA COMPLIANCE	TOOLING SUPPORT / COMMUNITY	PROJECT MIGRATION EFFORT	SOFTWARE INITIAL COST & SCALING COST	SOFTWARE YEARLY COST	VENDOR LOCK-IN POTENTIAL	Overall	Rank
1 IBM Websphere 6.1	Δ	Δ	Δ	Δ	X	X	Δ	X	3
2 Redhat JBOSS	Δ	Δ	O	Δ	O	Δ	Δ	Δ	2
3 Sun Glassfish 2.0	Δ	O	O	Δ	O	O	O	O	1

5. Proposed Solution

Proposal: Install Jetty in parallel with Websphere 6.1 and migrate in stages.

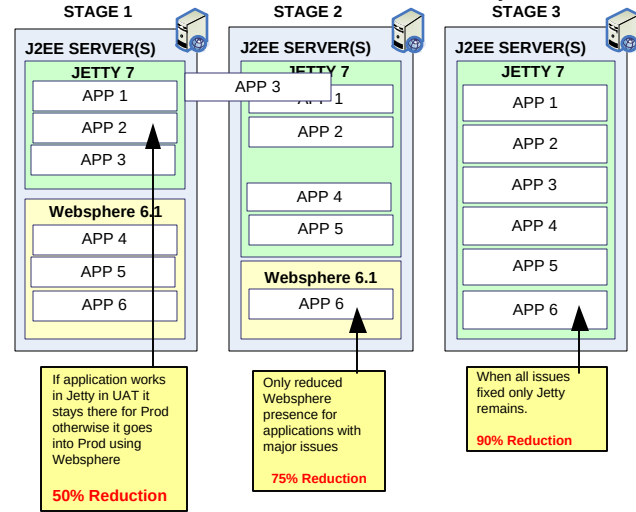


Migration: Test each application in Jetty in first instance

If tests are successful: Go live in Jetty.

If unsuccessful: Go live in Websphere 6.1 and then fix issue before re-deploy for go live in Jetty.

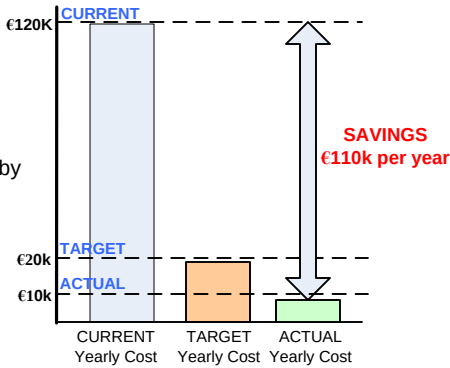
Staged deployment: At least 50% of apps will deploy without issue. Where Websphere still required we can fix issue and then move over time until 100% are live in Jetty



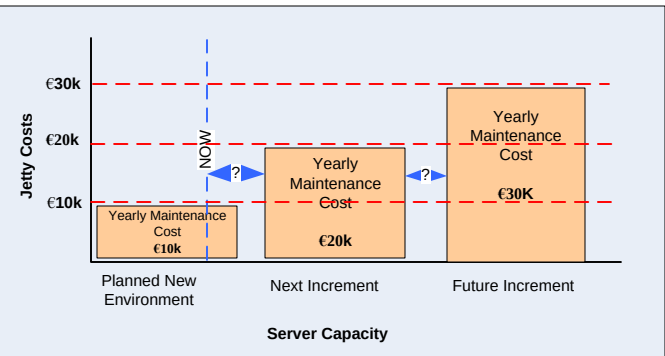
6. Benefits

Target Setting:

Massive yearly cost reduction achieved by changing platform.



Cost Scaling: Jetty maintenance costs scale per PROD server only



7. Additional benefits

Environment Consistency: Same J2EE application server used in very environment. Developers develop using it and then promoted through to Prod on same consistent version.

Developer Desktop

J2EE APP SERVER

Jetty

DEV J2EE Server

J2EE APP SERVER

Jetty

UAT J2EE Server

J2EE APP SERVER

Jetty

PROD J2EE Server

J2EE APP SERVER

Jetty

Other Notable Benefits:

- Reference Implementation of Java Language.
- Large community to learn from.
- Faster access to new java language updates
- Tight Integration to key frameworks used in TME applications (Spring etc)
- Better fit to hardware: Multi threaded architecture support.
- Simple file copy deployment mechanism, large reduction in effort required.

8 Investment

Breakdown of costs:

Jetty support: 3,660 per server = 7,320 per year

9)Implementation Risk and Mitigation

Applications with WAS specific functionality or not written to Java Standards may need some re-factoring / countermeasures.

This risk will occur regardless of Application Server chosen

Pilot done already on Jetty with NPPS apps show no issues.